

Midterm Assignment: Prompt Engineering

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Part 1: Prompt Comparison Analysis

- a. Domain and Task Description Domain Marketing Task: Write a “New Year’s Resolutions” email for a gym app targeting users who haven’t been to the gym in 3 months and encourage them to re-enter the app with a discount offer.
- b. Version A: Vague Prompt: Write an email yo users who have stopped logging into the app and mention our new year goal and offer a discount.

AI Response: Subject: We missed you! Return for New Year Goal!

Hello, We noticed you haven't used our app for a while. We are starting a New Year Goal (Challenge) soon and we think you will like it. It is a great way to get into shape. Because we want you back, we are giving you a discount code. Please use the code NEWYEAR at checkout. Hoping to see you again, **Fitness App Team**

c. Version B: Improved Prompt

Prompt: Role: You’re a persuasive email marketer for a premium fitness app and target audience (those who haven’t logged in for more 90 days). They’re likely feeling guilty about abandoning their fitness journey. **Task** Write an email for the 30 Day New Year Reset campaign.

Restriction:

- **Subject line:** Must be engaging but not exceed 50 characters.
- **Tone:** Encouraging, non-judgmental, and energetic.
- **Offer:** 50% discount on the annual plan using the code "RESET2025".
- **Length:** Under 150 words. **Structure:**
 1. Warm opening (acknowledge that life gets busy).
 2. "Hook" (Features of the New Year Reset).
 3. "Call to Action" (CTA) containing the discount. **Techniques Used:** Role Assignment, Specific Target Audience Definition, Tone Setting, Clear Constraints, Structured Output.

AI Response (Simulation): Subject: Ready for a reset?

Hi [Name],

We understand you—life gets in the way and routines can slip. But your fitness journey isn't over; it was just paused.

The **30-Day New Year Reset** is the perfect opportunity to get back. No pressure, just simple, daily wins designed to make you feel your best again.

Let's make 2025 your strongest year yet. Come back today and get **50% off** the annual plan.

[Get My 50% Discount - Code: RESET2025]

We're saving a spot for you! **Your Fitness Coach**

d. Comparison Analysis

Version B produced much higher quality output than Version A thanks to three key improvements I saw in the lessons:

1. Tone Control: Version A was too generic. Version B used tone features to address the psychological state of a user who had abandoned the app. This allowed us to connect more genuinely and beautifully with phrases like "life is getting in the way," as opposed to Version A's "We noticed you weren't using it."

2. Specificity and Constraints:

Version A had no restrictions, resulting in a standard subject line, but Version B used restrictions (under 150 words, subject line character limit), which forced the AI to be concise and compelling, which is important for marketing emails.

. Role Assignment: By assigning an "Email Marketing Specialist" persona, Version B was ensured access to a more professional vocabulary ("routines slip", "daily wins").

The structure provided in the prompt acted as an outline ensuring the logical flow (Empathy -> Hook -> CTA) was followed flawlessly.

Part 2: Applied Prompt Design

a. Framework Selection and Rationale Selected Framework: CO-STAR (Context, Objective, Style, Tone, Audience, Response) **Selected Task:** 1. Designing a customer onboarding email sequence.

Rationale: I chose the CO-STAR framework because email sequences require high precision in aligning tone and audience. Unlike a simple Q&A, an onboarding sequence must deliver a consistent narrative across multiple emails. CO-STAR separates "Context" from "Target" and clearly defines "Style" and "Target Audience," minimizing the risk of the AI appearing overly salesy or overly technical.

b. Full Prompt According to Framework Structure

[Context]: We are "TaskFlow", a new project management software for creative freelancers. Our unique selling point is visual simplicity. The user has signed up for a free trial but has not yet created their first project board. **[Objective]:** Write a 3-email onboarding sequence.

- Email 1: Welcome + Single core action (Create Project).
- Email 2 (Sent 2 days later): Tips for organizing visual boards.
- Email 3 (Sent 4 days later): Social proof (case study summary) + Nudge to upgrade to Pro. **[Style]:** Minimalist, punchy, and modern. Use short paragraphs and bullet points. Reference: "Written-planned mode" but with "Peer relationship". **[Tone]:** Enthusiastic, helpful, and inspiring. Avoid corporate jargon like "synergy" or "deliverables". **[Audience]:** Creative freelancers (designers, writers) overwhelmed by complex tools like Jira or Asana. They care about aesthetics and ease of use. **[Response Format]:** Output as a structured list. Provide the following for each email:

1. Subject Line
2. Body Text
3. Call to Action (CTA) Button Text.

c. AI Output (Simulated and Annotated)

Email 1 Subject: Welcome to the calmer way of working **[Aligned with Style: Minimalist/Modern]** **Body:** Hi [Name],

Welcome to TaskFlow! We are so excited to have you here.

You didn't choose to be a freelancer to spend hours managing spreadsheets. You chose it to create. That's why we built TaskFlow—to clear your path.

Ready to start? Your first visual board is waiting for you. Setup takes less than 60 seconds. **[Aligned with Objective: Single core action]**

CTA: [Create My First Board]

Email 2 Subject: Your tasks, but prettier **Body:** Hi [Name],

Now that you're settled in, let's make your workspace work for you.

3 ways to organize your TaskFlow board:

- **Color Codes:** Use tags for different clients.

- **Visual Uploads:** Drag and drop drafts directly onto cards.
- **Focus Mode:** Close the sidebar when you need deep work. **[Aligned with Audience: Creators needing focus]**

CTA: [Try Focus Mode]

Email 3 Subject: See how Sarah saved 10 hours a week **Body:** Hi [Name], Meet Sarah, a graphic designer drowning in sticky notes. Since switching to TaskFlow Pro, she gained back 10 hours a week for actual design work.

"TaskFlow is the only tool that thinks like a designer, not a project manager."

Unlock unlimited boards and client sharing features with Pro.

CTA: [Upgrade to Pro]

d. Refinement and Iteration

Identified Weaknesses:

1. **Lack of personalization:** The output uses "[Name]" but offers no suggestion on where we can insert different industry examples (e.g., "writing" instead of "design").
2. **Tone Shift in Email 3:** The 3rd email feels a bit too transactional compared to the "helpful/inspiring" tone requested in the CO-STAR prompt; it jumps to sales too quickly.

Improved Prompt (Iteration): Critique and Correction: The previous output was good, but Email 3 felt too aggressive. **New Instruction:** Rewrite Email 3 only. **Constraint:** Use a "Soft Sell" approach. Instead of asking them to upgrade immediately, ask how their first week went, then present the Pro feature as a solution to a problem (e.g., running out of space). **Format:** Add a "P.S." section to the email and put the upgrade offer there; so it feels like a personal check-in, not a sales email.

Improved Output

Email 3 (Revised) Subject: How is your first week going? **Body:** Hi [Name],

It's been a few days since you joined TaskFlow. How does the creative chaos feel? We hope it's a bit more organized!

We'd love to hear what you're working on. Reply and let us know if you need any tips on setting up your workflow.

Keep creating, **The TaskFlow Team**

P.S.: Running out of space on your free board? TaskFlow Pro offers unlimited boards and client sharing, so you never hit a wall. **[Compliant with "Soft Sell" Constraint]**

CTA: [Explore Pro Features]

Explanation: This correction used the **Iterative Refinement** principle. By specifically criticizing the "transactional" nature of the first output and applying a new constraint ("Soft Sell" + "P.S. Format"), the output transformed from a hard sales pitch into a relationship-building email that better aligned with the original "Empathic" tone request.

e. Framework Effectiveness Reflection

The CO-STAR framework was extremely effective for this task. Defining "Audience" (Creative Freelancers) and "Style" (Minimalist) separately prevented the AI from generating generic, corporate-sounding emails. However, I realized that for such a complex task, **Context** was key; I had to be very specific about what the software does.

If I were doing this again, I might consider the **RACI** framework if I were simulating a scenario where the AI acts as a "Junior Copywriter" submitting a draft to a "Senior Editor" (me). However, for pure content generation, CO-STAR is superior because it forces you to clearly define "Style" and "Tone", which are usually the first things to go wrong in creative writing tasks.

Part 3: Critical Reflection

a. Skill Development

My understanding of prompt engineering has evolved from viewing it simply as "asking questions" to "programming with natural language." Initially, I thought a good prompt just needed to be polite or grammatically correct. Now, I understand it as an engineering process where inputs (context, constraints) deterministically shape the outputs.

My "Aha moment" occurred in Week 2 when we learned about **tokenization**. Realizing that LLMs don't "read" words but instead process numerical tokens explained why they struggle with simple spelling or math operations when not given the correct command (e.g., "think step-by-step").

The most challenging concept was striking the balance between specificity and creativity. In Week 3, I fell into the mistake of over-constraining the model, resulting in dry outputs. I overcame this by using **Tone Constraints** instead of strict content rules; this allowed the model creative freedom within a specific "vibe."

b. Practical Application

Scenario: In my future Project Management career, I plan to use prompt engineering to summarize meeting transcripts and take action. **Framework:** I would use the **RACI Prompt Canvas**.

- **Role:** "Expert Project Manager."

- **Action:** "Extract action items and assign owners."
- **Context:** "Transcript of Q3 planning meeting."
- **Instruction:** "Format as a table with columns: Task, Owner, Deadline, Priority."

Challenges: The main challenge is the **Context Window** limit. If the transcript is too long, the model might "forget" earlier parts. I would solve this by **chunking** the text (breaking it into pieces) or using a model with a larger context window (like GPT-4-turbo or Claude 3).

c. Limitations and Ethics

Two critical limitations of LLMs are **Hallucinations** and **Knowledge Cutoffs**. As we learned in the "Catch the Hallucination" activity, LLMs can confidently state falsehoods (like citing non-existent court cases).

Ethical Consideration: In a project management scenario, sending meeting transcripts containing confidential client data or personal employee information to a public LLM is a major privacy violation.

Responsible Use:

To ensure responsible use, I would anonymize all data before entering the prompt (replacing names with [Employee A], companies with [Company X]). Furthermore, I would verify all "facts" generated by the AI against the original recording; regarding the AI not as the final authority, but as a drafter.

d. Goals for Second Half

In the second half of the course, I want to explore advanced guidance is complex problem solving to tackle logic-heavy tasks.

The skill gap I want to address is Systematic Assessment. I know Instruction A is better than Instruction B, but it's difficult to understand why, and I'd like to learn more concretely.